

CHEMISTRY | UNIT 03 | LESSON 03

Covalent Bonding and Molecular Compounds

Checkpoint Answer Sheet

Name: _____ Date: _____

MLA standards: MLA.CHEM.BON.01 and MLA.CHEM.BON.02

Directions: Complete this checkpoint independently after reviewing your lesson notes and corrected practice. Show electron sharing, formula meaning, and evidence-based reasoning.

1. Draw and label a Lewis model for H₂O. Show each shared electron pair.
2. Explain why H₂O is a molecule rather than an ionic formula unit.
3. The formula CO₂ contains one carbon atom and two oxygen atoms. Explain how the formula and bonding model communicate the same information.
4. Correct this misconception: "A molecular formula can always be reduced to the lowest whole-number ratio."
5. Use claim-evidence-reasoning to explain why atoms share electrons in a covalent bond.